

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 658 Randomized Algorithms — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Motwani & Raghavan — Randomized Algorithms	The primary source and still the standard. Modules 02–09 follow it. Library copy; the authors' lecture notes from the courses it grew out of are free.
Mitzenmacher & Upfal — Probability and Computing, 2nd edition	The better first book , and stronger on Modules 02, 03, and 09. Library copy; the authors' course pages carry free problem sets.
Alon & Spencer — The Probabilistic Method	Module 07's reference , and the book that made the method a subject. Library copy; Spencer's free lecture notes cover the core.
Cormode & Yi — Small Summaries for Big Data (free)	Module 05's reference, free in full — sketches and streaming algorithms with their error bounds, written for practitioners.
Levin & Peres — Markov Chains and Mixing Times (free PDF)	Module 09's reference, free from the authors — the definitive treatment of mixing, and chapters 1–7 are what this course needs.
Vadhan — Pseudorandomness (free)	Module 12's reference, free in full — expanders, extractors, and generators, which is where derandomisation actually lives.

Module-by-module map

Module 01 — Why Randomness

Resource	What it covers for this module
Mitzenmacher & Upfal — Probability and Computing, chapter 1	The three uses and the two kinds of algorithm , with the verifying-matrix-multiplication example that motivates §2.
Motwani & Raghavan — Randomized Algorithms, chapter 1	The Las Vegas / Monte Carlo distinction and the adversary framing of §1.
Crosby & Wallach — Denial of Service via Algorithmic Complexity Attacks (free)	§1's adversary use, as an attack — and the paper that caused language runtimes to randomise their hashing.
Schroeder, Pinheiro & Weber — DRAM Errors in the Wild (free)	§3's hardware error rates, measured at scale — which is what makes the argument quantitative rather than rhetorical.

Module 02 — Concentration

Resource	What it covers for this module
Mitzenmacher & Upfal — chapters 3–5	The ladder, derived — including the Markov-to-Chernoff derivation of §1's callout.
Motwani & Raghavan — chapters 3–4	The same material with more applications , and the martingale treatment of §2's fallbacks.
Boucheron, Lugosi & Massart — Concentration Inequalities	The comprehensive reference — McDiarmid, Azuma, and everything beyond. Library copy; the authors' survey papers are free.
Dubhashi & Panconesi — Concentration of Measure for the Analysis of Randomised Algorithms	§2's negative association and martingale material , aimed exactly at this course's needs. Library copy.

Module 03 — Balls and Bins

Resource	What it covers for this module
Mitzenmacher & Upfal — chapters 5 and 14	Balls and bins, the Poisson approximation, and the power of two choices — the reference for this module.
Azar, Broder, Karlin & Upfal — Balanced Allocations (free)	§2's result, in the original — the paper that introduced the power of two choices.
Mitzenmacher — The Power of Two Choices: A Survey (free)	§2 and §3, surveyed — including the applications and the variants.
Karger, Lehman et al. — Consistent Hashing (free)	§3's last row , and the virtual-node technique in its original setting.

Module 04 — Randomised Data Structures

Resource	What it covers for this module
Pugh — Skip Lists: A Probabilistic Alternative to Balanced Trees (free)	§1's structure, in the original — short, and the argument for the code-size benefit is made explicitly.
Seidel & Aragon — Randomized Search Trees (free)	§1's treaps , including the quicksort correspondence.
Carter & Wegman — Universal Classes of Hash Functions (free)	§2's definition , and the construction that made hashing analysable.
Pagh & Rodler — Cuckoo Hashing (free)	§3's scheme , with the load-factor threshold derived.

Module 05 — Sketches and Streams

Resource	What it covers for this module
Cormode & Yi — Small Summaries for Big Data (free)	Free in full, and the reference for this whole module — every sketch in §1's table, with its bound and its merge operation.
Cormode & Muthukrishnan — An Improved Data Stream Summary: The Count-Min Sketch (free)	§2's sketch, in the original , with the bound derived.
Flajolet et al. — HyperLogLog (free)	§3's method , including why the harmonic mean is the right combiner.
Vitter — Random Sampling with a Reservoir (free)	§3's reservoir sampling , with the uniformity proof and the faster variants.

Module 06 — Random Graphs

Resource	What it covers for this module
Mitzenmacher & Upfal — chapters 5 and 6	Random graphs and the moment methods of §2 , at the right level for this course.
Bollobás — Random Graphs	The comprehensive reference for §1's thresholds. Library copy.
Newman — Networks	§3's real networks , including why they differ from $G(n,p)$ and which models fit. Library copy; the author's lecture notes are free.
Decelle, Krzakala, Moore & Zdeborová — the detectability threshold (free)	§3's most striking result — the information-theoretic limit on community detection.

Module 07 — The Probabilistic Method

Resource	What it covers for this module
Alon & Spencer — The Probabilistic Method	The reference for this module — all the variants of §2 and the local lemma. Library copy; Spencer's free lecture notes cover the core.

Resource	What it covers for this module
Mitzenmacher & Upfal — chapter 6	The method at this course's level , with the MAX-CUT and MAX-3SAT examples worked.
Moser & Tardos — A Constructive Proof of the General Lovász Local Lemma (free)	§3's constructive result — short, and the entropy-compression argument is worth reading in full.
Tao — Moser's entropy compression argument (free blog post)	§3's proof technique explained , more accessibly than the paper.

Module 08 — Randomised Rounding

Resource	What it covers for this module
Williamson & Shmoys — chapters 1 and 5 (free PDF)	Randomised rounding, free in full — the set cover derivation of §1 and the feasibility methods of §2.
Raghavan & Thompson — Randomized Rounding (free)	The originating paper , in the routing setting of §4.
Gandhi, Khuller, Parthasarathy & Srinivasan — Dependent Rounding (free)	§3's technique , with the negative correlation proved.
Motwani & Raghavan — chapter 4	The same material from the randomised-algorithms side , with the Chernoff analysis explicit.

Module 09 — Markov Chains and Mixing

Resource	What it covers for this module
Levin & Peres — Markov Chains and Mixing Times (free PDF)	Free from the authors, and the definitive treatment — chapters 1–7 and 12–13 cover §1 and §2, including coupling.
Mitzenmacher & Upfal — chapters 7 and 12	Markov chains and MCMC at this course's level , with the Metropolis construction derived.
Gelman et al. — Bayesian Data Analysis, 3rd edition (free PDF)	§4's diagnostics, free in full — including the Gelman–Rubin statistic and the effective sample size, by their originators.
Jerrum & Sinclair — The Markov chain Monte Carlo method (free survey)	§3's counting-to-sampling connection , and the conductance method applied to real chains.

Module 10 — Dimension and Sampling

Resource	What it covers for this module
Blum, Hopcroft & Kannan — Foundations of Data Science (free PDF)	Free in full, and the best treatment of this module — JL, concentration of measure, LSH, and randomised linear algebra in one place.

Resource	What it covers for this module
Andoni & Indyk — Near-Optimal Hashing Algorithms for Approximate Nearest Neighbor (free)	§2's constructions and the amplification analysis.
Halko, Martinsson & Tropp — Finding Structure with Randomness (free)	§3's randomised matrix methods , with the error bounds and practical algorithms. The standard reference.
Leskovec, Rajaraman & Ullman — Mining of Massive Datasets (free PDF)	§2's MinHash and LSH from the practitioner's side , free in full — and the chapter is the clearest introduction available.

Module 11 — Randomness in Distributed Systems

Resource	What it covers for this module
Fischer, Lynch & Paterson — Impossibility of Distributed Consensus with One Faulty Process (free)	§2's impossibility result — short, and the bivalent-configuration argument is worth following.
Ben-Or — Another advantage of free choice (free)	§2's randomised escape , in three pages.
Demers et al. — Epidemic Algorithms for Replicated Database Maintenance (free)	§3's gossip protocols , in the paper that introduced them to systems.
Brooker — Exponential Backoff and Jitter (free, AWS Architecture Blog)	§1's jitter requirement, measured — with the thundering-herd effect demonstrated and the variants compared.

Module 12 — Derandomisation

Resource	What it covers for this module
Motwani & Raghavan — chapter 5	Conditional expectations and limited independence — the reference for §1 and §2.
Vadhan — Pseudorandomness (free)	Free in full, and the reference for §3 — expanders, extractors, and generators, with the derandomisation applications.
Hoory, Linial & Wigderson — Expander Graphs and Their Applications (free)	§3's expanders , surveyed thoroughly and readably.
Luby & Wigderson — Pairwise Independence and Derandomization (free)	§2's technique , with the constructions and the enumerate-all-seeds argument.

Module 13 — Using Randomness Honestly

Resource	What it covers for this module
Agarwal et al. — Deep RL at the Edge of the Statistical Precipice (free)	§1's reporting standard , developed for reinforcement learning and applicable unchanged here — with free tooling.

Resource	What it covers for this module
O'Neill — PCG, and the random number generator survey (free)	§2's generators compared , including the statistical tests and the predictability discussion.
Heninger et al. — Mining Your Ps and Qs (free)	§2's boot-entropy failure, measured in the wild — thousands of duplicate keys in deployed devices. The canonical example.
Brooker — the AWS architecture posts on jitter and timeouts (free)	Module 11 §1's practice, documented — and a model for how to report a randomised design decision.